

1) Choose a dataset

For our final project, we developed a visualization analyzing Olympic Dive Scores and Performance across multiple events and years. Because no, single, downloadable dataset existed that contained the level of detail required for our analysis, we constructed the dataset ourselves using publicly available competition results. We were specifically interested in round-by-round scores and how the type of dive performed and the difficulty of the dive impacted overall performance. This process required careful sourcing, manual data collection, and deliberate data cleaning decisions to ensure accuracy, consistency, and reproducibility. Initially we only found dive data on wikipedia. However, the primary source we ended up using for our data came from the official World Aquatics website, specifically from their Results Database (available here:

<https://www.worldaquatics.com/results?year=2026&month=latest&disciplines=>). This website provides detailed results on all international aquatic competitions, including all Olympic diving events.

The results database for dive competitions includes round-by-round dive scores and statistics. For the items of the results database, each row represents a diver. The attributes or columns of the database are: Overall Rank (final medal placement), Country (country represented), Athlete (diver first and last name), Age, Points (total points scored in the competition), Pts Behind (points behind first at the end of the competition). When you click the “+” dropdown to the right of the diver’s name you will see the following additional columns: Dive (dive round), Dive No. (code for type of dive performed), DD (degree of difficulty), J1-J7 (individual judges scores for all 7 judges), Dive Points (score for that round), Total Points (cumulative point total), Dive Rank (how the individual dive score compared to the other divers in that round), Ov Rank (overall rank in that round), Points Behind (points behind leader after each round). One challenge was that the World Aquatics database presents the data in an interactive web format and could not be easily exported into structured files like CSV or Excel. As a result, we manually extracted the relevant data by copying it from the website and entering it manually into multiple excel files that we later exported to CSV files. While this approach required additional effort, it allowed us to create a dataset tailored specifically to our project goals and interests.

We focused on the diving results from the 2016, 2020, and 2024 Olympic Games. We limited the scope of our dataset to focus on the four main individual events: Women’s 3m Springboard, Men’s 3m Springboard, Women’s 10m Platform, and Men’s 10m Platform. Initially, we considered including synchronized diving events, but we decided against this because it would introduce additional complexity, such as multiple athletes per entry, combined scoring, and different scoring criteria (i.e. performance and synchronization) – and would overall complicate our data’s structure and the visualization design.

First, we had to convert the excel files into csv files. Our final dataset was organized into 5 csv files, one for each event and one file that matched the dive_no (dive code) to the actual name of the dive performed. The 4 event csv files follow the same structure to ensure consistency across all events. For the items of the 4 event datasets, each row represents a diver. The attributes or columns are: final_rank (final placement), country, athlete (diver first and last name), age, points (total score), pts_behind (points behind first at the end of the competition), dive_no (dive performed in that round), dd (degree of difficulty of the dive in that round), J1-J7 (individual judges score for that round), dive_points (dive points earned in that round), total_points (cumulative score), dive_rank (how the individual dive score compared to the other divers in that

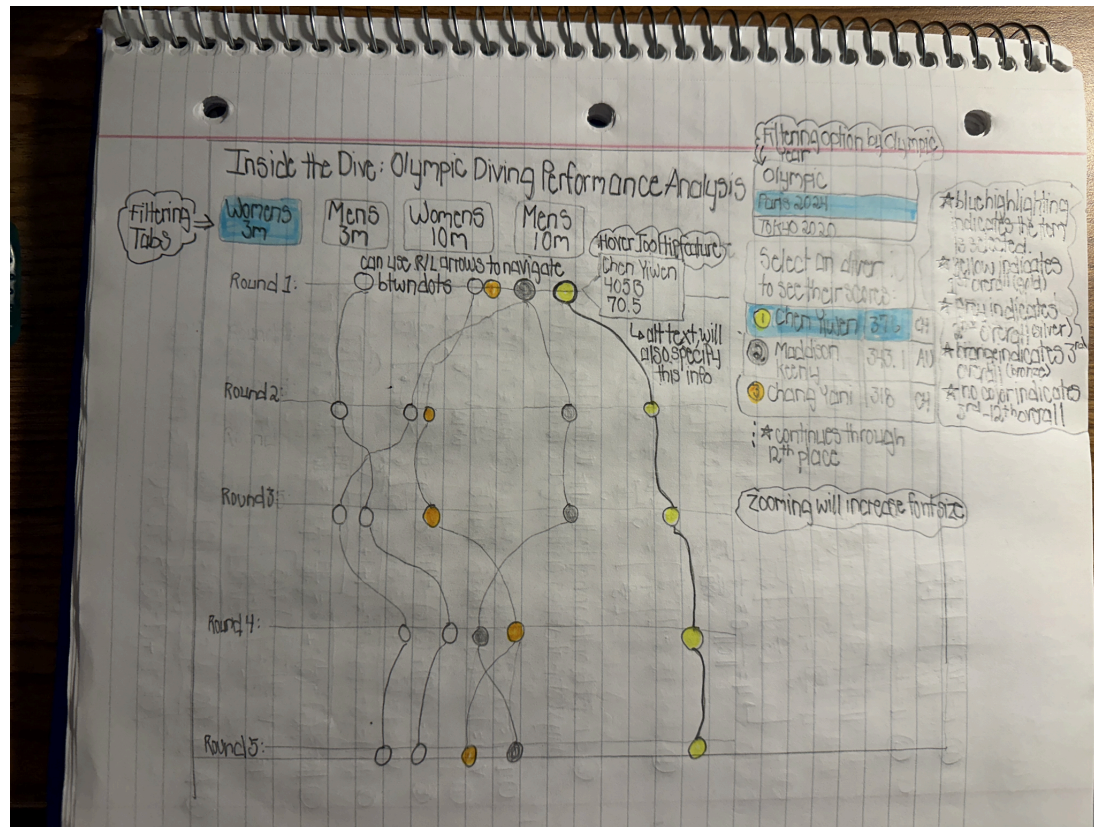
round), overall_rank (overall rank in that round), pts_behind (points behind after each round), Olympic_Year, Gender and Event. The round-by-round attributes were labeled by “R1_” through “R6_”. The additional csv file we created matches the dive code name to the actual names of the dive performed. For the items, each row represents a unique dive performed. The attributes or columns of the file are: dive_no and dive_name. We referred to the FINA degree of difficulty table (available here: <https://diving-gbdf.com/downloads/fina-dd.pdf>).

The data cleaning process involved several important steps. First, during the manual copying phase from the World Aquatics Results Database, we carefully verified that each value copied from the website was accurate. Because this step was prone to human error, we crosschecked entries, particularly paying attention to the degree of difficulty and score entries. We also performed basic excel functions to verify our data and standardized the layout and formatting across all sheets so that each event followed the exact same column order, naming conventions, data type, and overall format. One challenge in structuring the dataset was that not all divers complete the same number of dives in each event. In the Women’s event only 5 dives are performed, but in the Men’s event 6 dives are performed, which is why we separated the women’s and men’s event into different spreadsheets as we wanted to prevent having null data entries. Our dataset has no missing values. Next, we standardized formatting across the dataset to make any column with numeric values (i.e. score and degree of difficulty) numeric format. Text fields such as names, countries and dive categories were cleaned to remove extra spaces and ensure consistent capitalization and naming conventions. We also added the following derived attributes, Olympic_Year, Gender, and Event, to each row. Although this information is clear in the World Aquatic website’s database, we needed to include it in our customized dataset to allow for easier filtering and comparison across years for our visualization. No AI or GPT was used in the data cleaning process.

2) Paper Prototypes

As part of our visualization design process, we created multiple paper prototypes to explore different ways of representing the diving data, including annotations for interactivity, filtering, and accessibility. These sketches allowed us to experiment with different encoding techniques, interaction designs, and layout structures and emphasized different aspects of the dataset.

Sketch 1: Network Visualization



Sketch 1 Caption: Network visualization showing diver score progression across rounds, with items/nodes representing individual scores and links representing transitions between rounds.

This first prototype was inspired by:

https://public.tableau.com/app/profile/yvette/viz/WomenDivingResults_FINABudapest2017/Women1mSpringboard. In this design, each node represents a diver's score in a specific round, and the links connect those nodes across rounds to show how a diver's performance changes throughout the event.

This approach highlights performance trajectories and makes it easy to identify consistency and variability in individual round scores. At the top we included four tabs corresponding to each dive event. Selecting a tab dynamically updates the visualization. In the upper right corner, there is a filter dropdown that allows users to select a specific Olympic year. There is also a side table listing the final rankings for that event.

Sketch 1 Encodings and Design Choices:

- **Node position (vertical):** round number (round 1-5 for women's events and 1-6 for men's events)
- **Node position (horizontal):** individual dive score value
- **Color:** medalists (gold, silver, bronze) are highlighted above and all other divers will be the same neutral color

Sketch 1 Interactions:

- **Hover tooltip:** displays diver name, dive performed, degree of difficulty, and score for that round
- **Click on a link or node:** highlights that diver's full score progression

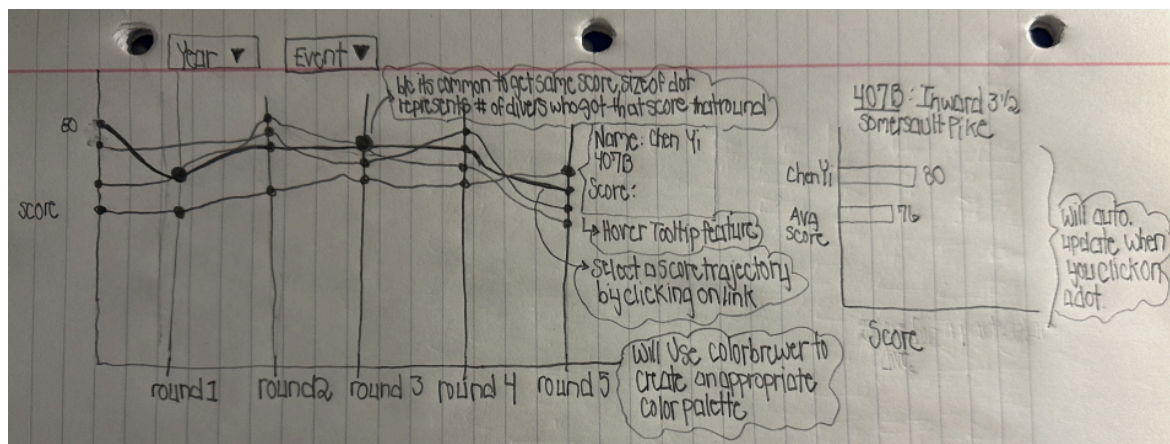
- Side panel selection: clicking a diver's name highlights their score path

A key design decision is that this visualization shows individual round scores only (not cumulative total score). This makes it easier to spot outliers, such as really strong dives or really bad dives. When multiple divers receive the same score in a round, they are represented by the same single node and clicking the node reveals all associated divers who received the same score).

Sketch 1 Accessibility Considerations:

- Color is supplemented with interaction to avoid reliance on color alone
- Tooltips provide detailed text descriptions
- You can tab between points using your arrow keys
- Potential zoom functionality would allow users to focus on regions of the graph where there may be scores close to each other

Sketch 2: Parallel Coordinates Visualization



Sketch 2 Caption: Parallel coordinates plot showing dive score distributions across rounds, with line paths representing individual divers.

This second prototype uses a parallel coordinate design. Here, the x-axis represents rounds, and the y-axis represents score values (non-cumulative). Each diver is represented as a line connecting their scores across rounds, allowing for easy comparison of patterns and trends.

Sketch 2 Encodings and Design Choices:

- X-axis: rounds (round 1-5 for women's events and 1-6 for men's events)
- Y-axis: score values
- Line: represents an individual diver's performance across rounds
- Dot size: indicates how many divers received the same score in a given round

Sketch 2 Interactions:

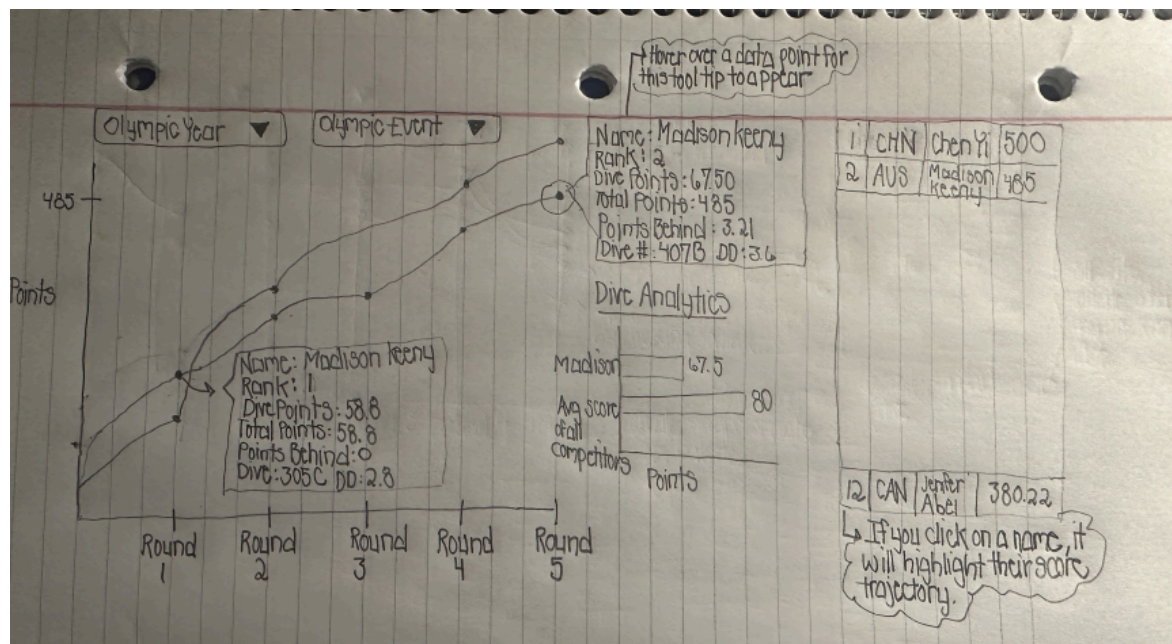
- Hover tooltip: shows diver name, dive type, and score
- Click on a line: highlights that diver's score path
- Click on a node (dot): triggers a sidebar visualization
- Filtering: dropdown menus allow user to filter by olympic year and event type

A unique feature of this prototype is the introduction of the linked sidebar chart, which appears when a user selects a diver or a specific score. This bar chart compares the diver's score on a given dive to the average score for that diver across all competitors in the same event and gender. This provides important context that is not immediately visible in the main chart.

Sketch 2 Accessibility Considerations:

- Larger nodes help indicate overlapping values
- Interactive highlighting reduces visual clutter
- More iterations could be adjustable to handle overplotting

Sketch 3: Line Chart Visualization



Sketch 3 Caption: Line chart showing both individual round scores and cumulative total scores, with interactive tooltips and comparison features.

This third prototype is a traditional line chart, but it incorporates both individual round scores and cumulative total scores. This allows users to understand not only how a diver performs in each round but also how those performances contribute to their overall standing. This design is partially inspired by: <https://nbremer.github.io/babynames/>.

Sketch 3 Encodings and Design Choices:

- X-axis: rounds
- Y-axis: point value (both per-round and cumulative)
- Line: represents a diver's progression
- Side table: displays diver name, country, final rank, and total score

Sketch 3 Interactions:

- Hover tooltip: shows detailed information including name, round rank, points scored, cumulative total, points behind first place, dive performed and degree of difficulty
- Click on a point: opens a bar chart comparing that diver's score on that dive to the average score of that dive against their competitors.

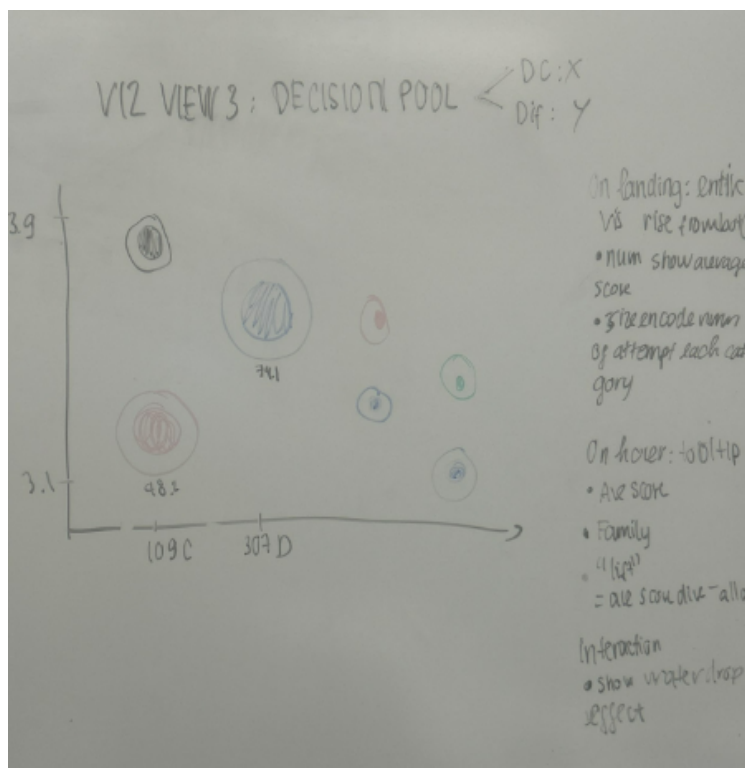
- Click on a name in the side table: highlights the diver's full trajectory
- Filtering: Tab dropdowns allow filtering by event and Olympic year

Compared to the first and second prototype, it is more straightforward but slightly less effective at showing distribution patterns across multiple divers simultaneously. We may also add another small bar chart showing the diver's individual round scores.

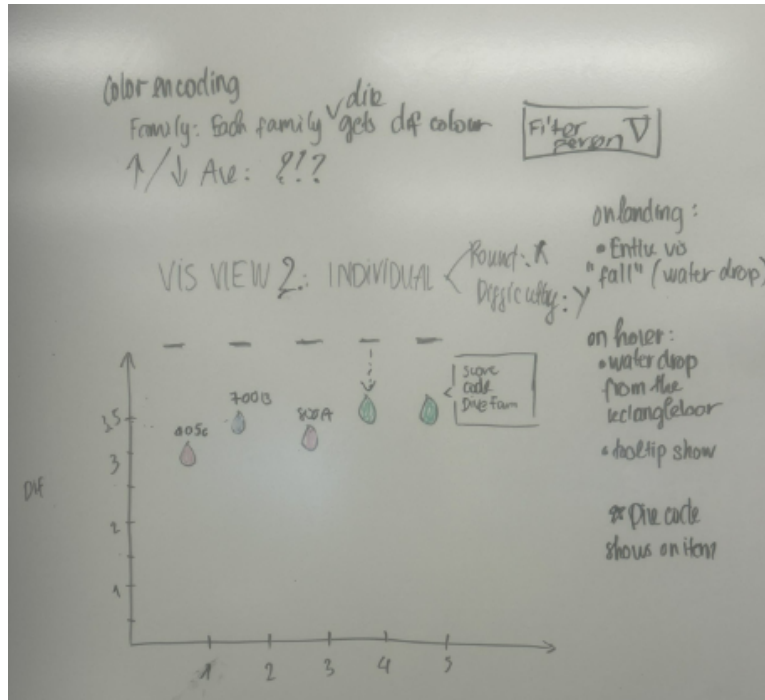
Sketch 3 Accessibility Considerations:

- Clear axis labels and structured layout improve readability
- Tooltips provide descriptive context
- The side table offers an alternative way to navigate the data

Additional Sketches



Additional sketch 1 caption: Bubble plot for decision pool comparing the degree of difficulty of a dive to the type of dive performed.



Additional sketch 2 caption: Scatterplot for an individual diver's dive difficulty for each round.

Iteration and Design Evolution

Across these prototypes, we explore different ways of encoding the same datasets. The network visualization emphasizes relationships and transitions, the parallel coordinates plot highlights distribution and comparison, and the line chart focuses on clarity and cumulative performance. During this process we identified several areas of improvement to note:

- The network visualization may become cluttered with many overlapping lines, suggesting a need for filtering or zooming
- The parallel coordinates may become overplotted
- The line chart, while clear, may not effectively show how individual performances compare across all divers per each round without additional context

3) Task Analysis + Accessibility Analysis

Below we describe each task, explain how a user would complete the task and we also include accessibility considerations.

Task Analysis

Task 1: Compare Values

Task 1: Between two divers (Matsuri Arai vs. Alejandro Orozco Loza), who has a higher cumulative score after Round 3 in the Women's 10m event in 2024?

How the user completes this task:

1. The user first selects the desired olympic year and event using the dropdown filters at the top of the visualization.
2. In the main line chart, each line (mark) represents a diver, and each point along the line corresponds to their cumulative score at a given round.
3. To compare 2 specific divers, the user can:
 - a. Either visually trace the diver's lines and compare the vertical position (y-axis) of the points at Round 3, or
 - b. Click each diver's name in the sidebar table, which will adjust the main line chart to only show the selected diver's score path.
4. The diver whose point is higher on the y-axis at Round 3 has the higher cumulative score.

Channels used:

- **Position (vertical/y-axis)** → encodes cumulative score
- **Line** → shows progression across rounds
- **Color** → distinguishes the different divers

Position is the most accurate visual channel for quantitative comparison, and isolating lines reduces clutter, making comparisons more precise.

Task 2: Determine Range

Task 2: What is the range of round 1 dive scores in the Men's 3m diving event in 2016?

How the user completes this task:

1. The user selects a specific year and event using the dropdown filters.
2. The user then can hover over the highest first point (the top-most point on the y-axis) and the lowest first point (bottom-most position) to see the highest and lowest score for round 1 (They could also select the two purple line divers to compare)

Channels used:

- **Position (vertical/y-axis)** → encodes score magnitude
- **Text (sidebar table)** → provides exact values

The line chart allows for quick visual estimation of the range, while hovering allows you to see the exact score for a specific round.

Task 3: Order Items

Task 3: Order the divers from lowest to highest total score in the Women's 3m event in 2020.

How the user completes this task:

1. After selecting the year and event using the filter dropdowns, the user looks at the right sidebar table to see the final rankings.
2. The table already lists divers in the ranked order (from first to last), including additional information such as diver name, country, and final score.
3. The user can read the list from bottom to top (lowest to highest) or top to bottom (highest to lowest), depending on the desired ordering.

Channels used:

1. **Text and Table Position** → encodes ranking order

2. **Numeric labels** → provide exact total scores

Ordering is most effectively supported through the ranked sidebar table, as it provides the exact ordered ranking without requiring the user to guess.

Accessibility Analysis

Based on the WAVE accessibility guidelines, we refined our accessibility approach to ensure that our interactive Olympic diving visualization is compatible with screen readers and keyboard navigation. **Our visualization consists of:**

1. A **line chart (SVG)** showing cumulative scores across rounds
2. **Dropdown filters** (for olympic year and event)
3. A **sidebar table** with rankings
4. A **linked bar chart** for individual dive score vs. avg dive score

To make this accessible, we focused on:

1. Meaningful HTML structure (specifically with headings, tables, and labels)
2. Alt text (hidden + visible captions)
3. Keyboard navigation (tabbing + selection)
4. Text-based equivalents for visual tasks

Task 1: Compare Values

A screen reader can announce structured data such as:

- “Diver A, Round 3 cumulative score: 250”
- “Diver B, Round 3 cumulative score: 270”

The user compares the numerical values directly to determine which is higher.

Users can:

- Navigate to the sidebar table, which lists divers and final scores
- Use keyboard navigation (tab key) to select a diver
- Hear values read aloud, such as: “Hongchan Quan, final score 425.6”
- For round-by-round comparisons:
 - Each SVG point (mark) includes the tabindex=“0” function and an aria-label. This allows users to:
 - Tab through each point
 - Hear exact values instead of relying on position
- **Limitation:** Users lose the ability to quickly compare many divers at once (i.e. there is a visual scanning advantage).

Marks and Channels (non-visual):

- **Visual points** → represented as text entries
- **Position** → replaced by explicit numeric values

Trade-offs:

- Forces precision (as it reports exact values)
- Loses quick visual comparison across many divers

Task 2: Range

For a screenreader, we provide a text summary near the chart specifying the data analytics (e.g. “Final scores range from 250.35 to 420.6”). Additionally, the sidebar table allows users to find minimum and maximum values manually and the table is structured with <table> tags for accessibility.

Marks → Text translation:

- **Visual points** → replaced by explicit minimum and maximum values, textual summary and ordered values

Limitation: It's efficient if summaries are provided, but without summaries, identifying range would require scanning all entries sequentially.

Task 3: Order Items

The sidebar table is already well-suited for accessibility. A screen reader can read it (e.g. “Rank 1: Honchan Quan, 425.60 points”) and users can navigate the table in order or sort if needed.

Marks and channels:

- **Visual ordering** → replaced by sequential text structure

Trade-offs:

- High clarity and precision
- Less contextual understanding of performance trends

Additional Accessibility Features

To further improve accessibility, the visualization includes:

- **Alt text** descriptions summarizing the chart (e.g. “Line chart showing cumulative diving scores increasing across rounds for 12 divers”). Our alt text summarizes the key takeaway of the visualization, which ensures general users get the main takeaway and that screenreader users get a more detailed explanation of functionality.
- **Keyboard navigation** for dropdowns, sidebar selection, and data points
- **Screen reader-compatible tooltips** that announce dive details and comparisons
- **Text summaries** for the comparison bar chart (e.g. “This dive scored 75, above the average of 68”).
- **Colorblind-friendly design**, ensuring color is not the only distinguishable feature

4) Pilot the tasks and (mostly working) vis

In this pilot version of our visualization, beneath the main line chart, a “Decision Pool” bubble graph provided additional context, highlighting overall event insights such as the most frequently performed dives and their associated difficulty levels. Below is the exact script we used during pilot testing:

Speaker Script

Speaker: “We are evaluating our visualization and are asking you, the participant, to complete some tasks using the visualization and then provide feedback about the visualization and

experience. As a reminder, we are evaluating the visualization, not you as a participant, so you don't need to worry about being "right" as you complete these tasks. There are three tasks, followed by a brief feedback session. The whole pilot session should take under 5 minutes. Do you consent to participate?"

[Wait for yes]

Speaker: "Thank you for agreeing to participate. We will start with the three tasks. Please 'think aloud' as you complete the task, meaning voice what you are thinking as you work through the task. **Your first task is:** Between two divers (Matsuri Arai vs. Alejandro Orozco Loza), who has a higher running total after Round 3 in the Women's 10m event in 2024?"

Note: "You may ask clarifying questions, such as "what do you mean by x?" or "are you looking for an exact number or an approximation?"

[Pause to allow the participant to complete the task]

Speaker: "Your second task is: What is the range of round 1 dive scores in the Men's 3m diving event in 2016?"

[Pause to allow the participant to complete the task]

Speaker: "Your third task is: Order the divers from lowest to highest total score in the Women's 3m event in 2020."

[Pause to allow the participant to complete the task]

Speaker: "That is the end of the third and final task. For this last bit, we welcome any feedback you may have about the visualization or about your process for completing the tasks."

[Allow participant to speak first, then informal discussion]

Note-taker Observations

Notes on how the participant completed each of the 3 tasks

Participant: (In-class on 5/4)

1. **Task 1:** The participant initially struggled to understand the cumulative score round by round because the accumulation was not explicitly visualized in an intuitive way. Despite confusion, the participant was able to complete the task.
2. **Task 2:** The participant was able to identify the range of values; however, did not immediately notice the section of the interface intended to highlight or support this task.

3. **Task 3:** The participant was able to complete this task easily, primarily due to the clarity and structure of the rankings table.

Feedback from the participant

Participant Feedback:

- The participant had difficulty locating and interpreting the comparison feature. They were unsure what the displayed numbers represented.
- The instructions provided were not sufficient for the participant to fully understand how to use the comparison functionality.
- The participant struggled to recognize that the comparison feature only reflects a specific round.
- The visualization was too text heavy so they found it somewhat overwhelming at first.
- Distinguishing between cumulative score and total score was particularly confusing.

List of changes that we plan to make for our final vis

- **The current “decision pool” visualization was confusing to users.** Revise or remove the decision pool visualization and provide clearer explanations of what the values represent.
- **Improve instructions and labeling:** Add clearer, more prominent guidance to help users understand how to use interactive features, especially when comparing divers.
- **Clarify score meaning:** Provide additional context or annotations to help users better understand how diving scores work and what they represent.
- **Reorganize layout:** Move the final scores and rankings higher on the page to make them more immediately visible to access.
- **Reduce text/cognitive overload:** Simplify the interface where possible and rely more on intuitive visual encodings rather than dense text explanations.
- Provide the user with an option to learn more about how dive scoring works, dive codes, dive performance categories and degree of difficulty.

5) User testing

After initial pilot testing, we used comments and feedback to finalize our visualization. We used the same script as we did in our pilot testing. Below are the results:

Results

Participant 1: Ann Healy

- **major/minor:** Political Science
- **Performance Notes:**
 - **Task 1:**
 - The participant initially struggled to interpret the visualization, particularly when trying to identify results by round.
 - Understanding final standings was also difficult (e.g. comparing athletes)

- The decision pool visualization was confusing and the line chart was not immediately intuitive
 - After some exploration and interaction with the visual, the participant was eventually able to locate the round 3 scores.
 - **Task 2:**
 - Filtering was straightforward and intuitive. However, interpreting the dot plot was not immediately clear, and the participant needed to zoom in to better understand the data. After doing so, they were able to complete the task.
 - **Task 3:**
 - This task was completed quickly and without difficulty. The participant was able to use the table, filter and identify the correct information successfully.
- **Feedback:**
 - Once the participant understood the interface, the visualization became easy to use and visually more clear.
 - The amount of text and multiple visual elements made it difficult to quickly interpret the information.

Participant 2: Lily Dyslin

- **major/minor:** Economics (major)/ Data Science (minor)
- **Performance Notes:**
 - **Note:** Due to poor results in previous testing cases, substantial additional changes were made before participant 2 and participant 3 user testing such as the removal of the decision pool visualization, change in formatting/style, less text, and clearer user instructions.
 - **Task 1:**
 - The participant had difficulty locating the correct individuals within the current layout.
 - A search function for names may improve usability, as scrolling through the list was inefficient. It took some time to locate the correct information.
 - However, much of this was due to the task question not being directly in front of the user. Once the user could read the task themselves and refer back to it, the participant was able to complete the task successfully.
 - **Task 2:**
 - The participant completed this task quickly and with minimal difficulty.
 - **Task 3:**
 - This task was completed quickly and without difficulty. The participant was able to use the table, filter and identify the correct information successfully.
- **Feedback:**
 - Overall, there appears to be a learning curve associated with interpreting the visualization. However, the interface is generally user-friendly once understood.
 - A key challenge was interpreting the task questions clearly.
 - Based on feedback we removed the decision pool graph to improve clarity and saw the participant performed better without it.

Participant 3: Aiyanna Siew

- **major/minor:** Chemistry
- **Performance Notes:**
 - **Task 1:**

- The participant focused on the bottom of the visualization where the dive rounds are labeled and correctly identified round 3.
 - She attempted to determine the highest score by examining the line chart with at 12 divers showing and found the chart difficult to interpret due to overlapping lines.
 - She did not listen or read the question carefully resulting in some confusion and delay.
 - She did not read the provided instructions on the website for how to navigate the visualization.
 - She noticed the different colors of the line (e.g. identified Alejandro as yellow line), she did not intuitively use the sidebar table.
 - Even after prompting, she did not immediately recognize the presence of the filters at the top of the visualization. However, after some time, she was able to complete the task.
- **Task 2:**
 - After becoming familiar with the filters, she was able to use them effectively.
 - She quickly navigated to round 1 and completed the task with minimal difficulty.
- **Task 3:**
 - She was able to apply the filters and reference the table efficiently.
 - She correctly assumed that the final score represented the total and successfully identified both the highest and lowest scoring divers.
- **Feedback:**
 - The participant noted that she is not very comfortable with technology, which contributed to her initial difficulty in noticing and using the filters.
 - Recommended to explicitly note that the sidebar table can be used to focus on individual divers.
 - Suggested making the filters more prominent, with clearer visual signifiers.
 - Adding a more explicitly filter heading
 - Move the “Show Diving Insights” and “View Underlying Data” buttons

Summary on Participant’s Performance

Overall, user testing revealed a mix of strength and many areas for improvement in the visualization, particularly related to clarity of instructions, layout, and the discoverability of interactive features.

One of the main areas for improvement was usability, especially during the initial interaction with the visualization. In Task 1, several participants struggled to interpret the line chart due to overlapping lines and did not immediately recognize key features such as filters or the sidebar table. A consistent issue across users was that some did not fully read or follow the provided instructions, which contributed to confusion and slower task completion. Additionally, important interactive elements—such as the bar chart and filters—were not always immediately noticeable, limiting users’ ability to efficiently navigate the visualization. Despite these challenges, there were several clear positives. Once users became familiar with the interface, performance improved significantly.

In Tasks 2 and 3, participants were generally able to use filters effectively, interpret the table, and correctly answer questions with greater speed and accuracy. The visualization's interactivity, particularly the filtering system and linked views, proved to be intuitive after initial exploration. Users were also able to correctly interpret key metrics, such as identifying highest and lowest scores, suggesting that the underlying data representation was sound.

To address the issues identified during testing, several targeted improvements were implemented. Instructions were revised to be shorter, clearer, and more explicit, with specific guidance such as directing users to scroll down to view the bar chart and explaining how the sidebar table can be used to isolate individual divers. The filter section was made more prominent by increasing its visual emphasis and adding a clearer heading. The overall layout was reorganized to improve flow and hierarchy, including repositioning buttons to more intuitive locations. Reducing excess text also helped streamline the user experience and minimize cognitive load.

After implementing these changes, the final version of the visualization was informally tested with additional users (friends). These follow-up tests showed noticeably improved results: users were able to navigate the interface more easily, identify key features more quickly, and complete tasks with fewer errors and less confusion. This iterative process demonstrated that enhancing clarity, simplifying instructions, and refining layout design had a meaningful impact on overall usability and user performance.

6) Finalize your visualization

At the top of the page users can filter the data using two dropdown menus (Olympic year and Olympic event), which dynamically update the visualization. There is also a View Underlying Data Button that shows the data in a table format and that also dynamically updates based on the filtering. The visualization has a main interactive line chart that displays cumulative diving scores across each round. An interactive sidebar table complements the chart by listing all divers who competed in the selected event as well as their final rankings, name, country they represented, and final total score, ordered from first to last place. Below the sidebar table there is Show Diving Insights Button and View Underlying Data Button to learn more about how dive scoring works. The visualization includes many accessibility features such as keyboard navigation (tabbing), uses a color-blind-friendly design, includes descriptive alt text, passes WAVE, and adheres to the POUR principles.

A final note on the visualization: While the visualization incorporates many accessibility features and was evaluated using the WAVE Web Accessibility Evaluation Tool, there is still one limitation in the line chart design. When displaying all 12 divers simultaneously, one color (dark purple) had to be reused. This is because after using ColorBrewer and searching other color-blind-friendly palettes, we could not find a color-blind-friendly palette that support 12 categories. It is important to note that the colors themselves do not encode a specific meaning. For the color that is repeated (dark purple), to maintain some visual structure, the first and last place divers lines in the line chart are both highlighted in dark purple to help users which could help the user understand the range of performance within the chart.

Link to Jordan's final vis: <https://jowhiteh22.github.io/>

Sources Used

(Jordan) Sources used to assist:

- Index.html sheet references:
 - https://www.w3schools.com/TAgs/tag_section.asp
 - https://www.w3schools.com/tags/tag_select.asp
 - https://www.w3schools.com/html/html5_semantic_elements.asp
 - https://www.w3schools.com/Tags/tag_div.asp
 - https://www.w3schools.com/accessibility/accessibility_labels.php
 - https://www.w3schools.com/tags/tag_label.asp
- Style.css sheet references:
 - https://www.w3schools.com/csS/css3_box-sizing.asp
 - <https://www.w3docs.com/snippets/css/why-and-how-the-bootstrap-sr-only-class-is-used.html>
 - https://www.w3schools.com/css/css_font_size.asp
 - https://www.w3schools.com/html/html_tables.asp
 - https://www.w3schools.com/cssref/atrule_media.php
 - https://www.w3schools.com/cssref/pr_grid-template-columns.php
- Main.js sheet references:
 - <https://www.tutorialsteacher.com/d3js/loading-data-from-file-in-d3js>
 - <https://d3js.org/d3-fetch>
 - https://d3-graph-gallery.com/graph/basic_datamanipulation.html
 - <https://d3js.org/d3-fetch>
 - <https://observablehq.com/@observablehq/introduction-to-promises>
 - <https://www.freecodecamp.org/news/flat-and-flatmap-javascript-array-methods/>
 - https://d3-graph-gallery.com/graph/interactivity_tooltip.html
 - https://www.w3schools.com/howto/howto_css_tooltip.asp
 - https://www.w3schools.com/howto/howto_css_display_element_hover.asp
 - <https://www.geeksforgeeks.org/javascript/how-to-add-a-tooltip-to-a-div-using-javascript/>
 - <https://d3js.org/d3-selection/modifying>
 -
- General:
 - <http://nbcolympics.com/news/diving-101-olympic-scoring-rules-and-regulations>
 - <https://htmlcolorcodes.com/>
 - <https://www.geeksforgeeks.org/css/css-cheat-sheet-a-basic-guide-to-css/>
 - https://www.w3schools.com/css/css_website_layout.asp
 - https://www.w3schools.com/css/css3_borders.asp
 - <https://www.geeksforgeeks.org/css/modern-css-features-you-need-to-know-in-2024/>

(Han) Sources used to assist (I had w3school opened and just searched for what I needed the entire time, along with the d3 library and general google searches, not sure if I needed to record them). I'm currently recording things I spent significant time looking at:

1. **Design and field constraint (All of this happened before code and after sketch)**
 - I used these sources to learn make sure I constraint my code to be adjustable for keyboard navigation

- <https://www.geeksforgeeks.org/javascript/javascript-addeventlistener-with-examples/>
- <https://www.youtube.com/watch?v=Q3ktcptd2yl>
- https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Classes/Public_class_fields#:~:text=cannot%20be%20constructor%20.,Description,once%20C%20at%20class%20definition%20time. → field usage
- <https://stackoverflow.com/questions/17883692/how-to-set-time-delay-in-javascript>
- Ensure dropdown compatibility
 - <https://developer.mozilla.org/en-US/docs/Web/HTML/Reference/Elements/option>
- Some websites I inspected to see whether I should implement constant fields or queue callstacks for my amount of data, and if so whether I should query. Ended up going with constant field.
 - <https://informationisbeautiful.net/visualizations/common-mythconceptions/>
 - <https://arborscope.com/mapDisplay.cfm?id=0FCEC3C6E0>
 -
- Learned some traversal, compatible data structs
 - <https://www.youtube.com/watch?v=v7rSSy8CaYE>
- DOM setup
 - https://developer.mozilla.org/en-US/docs/Web/API/Document_Object_Model, the entire thing
- Considered making program dynamic (can encode if more csv arrives from future Olympics, changed my mind)
 - <https://catstat-prediction-idv25qg5c5fxezia6yfdbq.streamlit.app/>
- Key Usage:
 - https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/Object/keys
- Data prep
 - https://www.w3schools.com/js/js_maps.asp
 - https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/Array/flatMap
 - <https://stackoverflow.com/questions/67353925/how-to-send-a-state-into-child-component-as-dataset-of-d3-js-object> minor state object debugging

2. Designing structs

- Revisited our d3 labs – had the [d3 library](#) open the entire time
- Revisited some of my outside of class works
- <https://codesignal.com/learn/courses/parsing-table-data-1/lessons/parsing-csv-files-with-javascript> parsing syntax pick-up
- dom lib
 - https://www.w3schools.com/js/js_htmlDOM_document.asp
 -
- General js objects
 - https://www.w3schools.com/js/js_object_definition.asp
 - https://www.w3schools.com/js/js_typed_reference.asp
 - [https://javascript.info/map-set#:~:text=new%20Set\(%5Biterable%5D\),size%20%E2%80%93%20is%20the%20elements%20count](https://javascript.info/map-set#:~:text=new%20Set(%5Biterable%5D),size%20%E2%80%93%20is%20the%20elements%20count).
 - <https://dev.to/anoop-rajoriya/map-and-set-in-javascript-4kc>
 - https://www.youtube.com/watch?v=SmJNeJuLmVo&list=PLZPZq0r_RZOO1zkqO4bldfuLpizCeHYKv&index=26
- String query
 - https://www.w3schools.com/js/jquery_selectors.asp
- Athlete grouping: [basically a rollup based grouping](#)

- Predominantly set and map setup (helps a little with deselection)
- More on d3 constructors: <https://www.youtube.com/watch?v=VPrrPEa0XDM>
- Dynamic chart size: <https://www.youtube.com/watch?v=XPOSEf40SkQ>
- Needed a lot of help with the circle packing diagram
 - <https://www.youtube.com/watch?v=QHEQuoIKgNE&t=47s>
 - <https://www.youtube.com/watch?v=YuJkT2gF7cc&t=1s>
 - <https://www.anychart.com/blog/2022/09/08/circle-packing-chart/> → did not help
 - Sticky header: <https://stackoverflow.com/questions/18382496/sticky-header-after-scrolling-down>

3. Formatting decisions

- <https://stackoverflow.com/questions/78639703/display-different-text-for-html-select-option-than-label>
- https://www.sussex.ac.uk/tel/resource/tel_website/accessiblecontrast/?q=FFFFFFF~003b49~1d4289~94a596~e56db1~d3273e~00bfb2~d6d2c4~ffc845~dc582a~41b6e6~1b365d~be84a3~5d3754~7da1c4~f2c75c~d0d3d4~007a78~000000 → color palette choice, didn't want to spend too much time on this so just found this open source WCAG comp palette and implemented it

4. Implementation control flow

- <https://www.youtube.com/watch?v=JloLGV9DmtQ> Watched this with lots of skip on 2x speed just to get some conditional syntax
- <https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Operators> some conditional syntax
- Delay syntax
 - https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/Promise
- <https://website.education.wisc.edu/~swu28/d3t/concept.html> some more d3 utilities

5. Program Revision 1

- https://developer.mozilla.org/en-US/docs/Web/Accessibility/Guides/Seizure_disorders

6. Program Revision 2: Keyboard Navigation

- <https://stackoverflow.com/questions/33005267/simple-html5-keyboard-navigation-for-elements>
- https://developer.mozilla.org/en-US/docs/Web/Accessibility/Guides/Keyboard-navigable_JavaScript_widgets
- <https://www.bennadel.com/blog/4667-using-both-tab-and-arrow-keys-for-keyboard-navigation.htm>
- Complex to learn and set up, but pretty easy to implement